

Visual Architecture & Diagrams Guide

Complete End-to-End System Topologies, UI Workplaces & Customizing Sequences

DOCUMENT CONTROL & SYSTEM SCOPE

- **Document Classification:** Architecture, System Topology & UI Reference
- **Document Version:** 2.1 (Production Release)
- **System Namespace:** /CTDI/
- **Graphic Specification:** Scalable Vector Graphics (Pure SVG 1.1 / OpenXML Vector Media)
- **Embedded Diagrams:** 8 Lossless Architecture & Workflow Diagrams
- **Target Roles:** Solution Architects, SAP Technical Leads, Key Users, Supervisors
- **Publication Date:** September 2026

Executive Summary & Visual Index

This visual guide consolidates the end-to-end architecture, user interface layouts, print spool workflows, and customizing lookup hierarchies of the **/CTDI/ Repair Print Framework**. All diagrams in this document are generated as lossless Scalable Vector Graphics (SVG) with pixel-exact font rendering, vector icons, and coordinate precision.

Fig #	Diagram Title	File	Architectural Scope
1	Workshop Repair & Print Life Cycle	images/workshop_1_lifecycle.svg	End-to-end service lifecycle from goods receipt to customer dispatch.
2	Single Order Selection Interface	images/ui_single_selection.svg	Operator interface for on-demand single repair slip generation.
3	Quick-Access Customizing Toolbar	images/ui_customizing_toolbar.svg	Direct maintenance shortcuts for key users on the selection screen.
4	Mass Processing Selection Workplace	images/ui_mass_selection.svg	Batch order filtering, plant/work center scoping, and serial range inputs.
5	Interactive ALV Workplace Grid	images/ui_alv_grid.svg	Multi-order operations cockpit, traffic light indicators, and 1-click drilldowns.
6	Spool Dispatching & Bundling Modes	images/spool_modes.svg	Comparative topology of Individual, Bundled, and Merged spool jobs.
7	Inspection Photos Multi-Slot Layout	images/image_layout.svg	DIN A4 coordinate calculation and aspect-ratio preservation for GOS photos.
8	Form Routing & Priority Resolution	images/access_sequence.svg	5-tier access sequence for customer/plant layout determination.

1. Workshop Repair & Print Life Cycle

The following diagram illustrates how the print framework coordinates between SAP PM/CS orders, technician workbenches, inspection photo repositories, and the physical print queues.

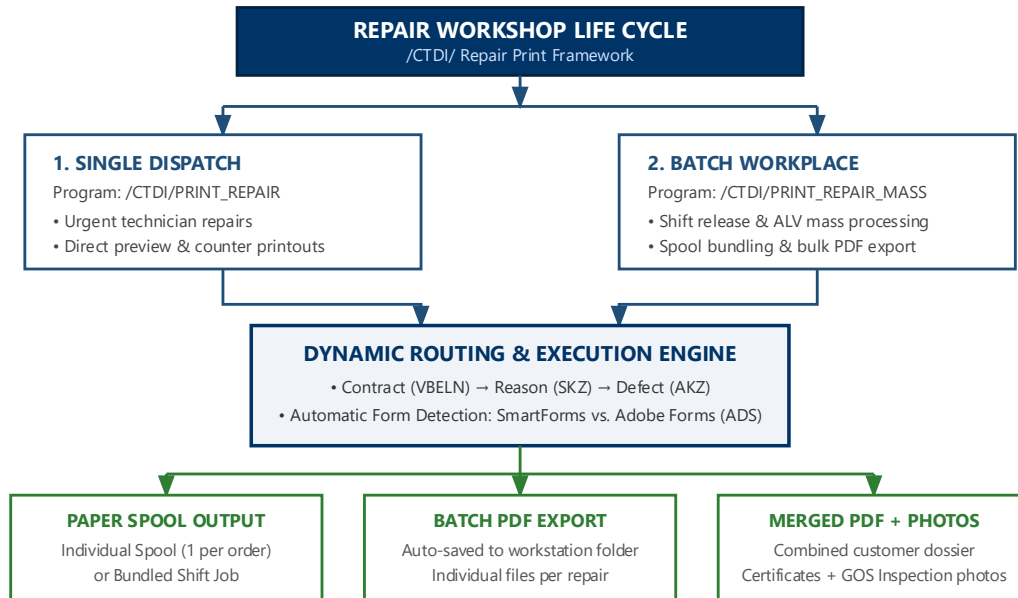


Figure: Repair Workshop Life Cycle

Architectural Phase Breakdown

1. Step 1: Receipt & Defect Logging ('IW51'/'IW31')

- The defective customer unit arrives at the receiving bay.
- Operators register the equipment serial number, log warranty status, and create the SAP Service Order (**AUFNR**).
- High-resolution photographic evidence is captured (damage inspection, packing state) and attached via Generic Object Services (GOS) or ArchiveLink.

2. Step 2: Technician Workbench & Diagnosis ('IW32'/'IW42')

- The workshop technician performs physical inspection, component replacement, and functional testing.
- Confirmations and spare part consumptions are posted.
- Outcome codes (e.g., *BER - Beyond Economical Repair*, *SCRAP*, *WARRANTY REPAIR*) are recorded.

3. Step 3: Document Rendering & Routing ('/CTDI/PRINT_REPAIR')

- The print framework interrogates the form routing table (**/CTDI/REP_FORMS**).
- It determines whether to execute an **Adobe Interactive Form (ADS)** or **SmartForm**.
- If GOS attachments exist, the image engine fetches the raw binaries, validates headers, and scales each image onto DIN A4 appendix pages.

4. Step 4: Dispatch & Customer Delivery

- Documents are dispatched to physical barcode printers, customer PDF archives, or local workstation drives.
- The repaired unit and the printed repair report are packed and routed to shipping.

2. Single Order Selection Interface

The single-order printing transaction `/CTDI/PRINT_REPAIR` provides an intuitive, high-speed interface for bench technicians and packaging personnel.

/CTDI/PRINT_REPAIR — Single Repair Order Printout (Interactive Dispatch)

▶Execute (F8) | Project (FC02) | Forms (FC03) | Results (FC04) | ⚡ Mass Printing (FC05)

Selection Criteria

Repair / Order ID: * 4001234 **Obligatory** SAP PM/CS Order Number (leading zeros auto-added)

Serial Number: SN-987654321 **Optional** Leave blank to print order master equipment

Additional Options & Attachments

☒ **Append attached images to PDF (GOS & ArchiveLink Content Server)**
Automatically extracts defect photos, scales to A4 with aspect ratio preservation, and appends to the final PDF.

● Ready | System: PRD | Output: Spool Default (IW42) / Local File Prompt (Manual) SAP GUI / CTDI Repair Framework

Figure: Single Repair Order Printout Selection Screen

Key Parameters & Logic

- **Repair / Order ID (`p_aufnr``):** The primary 12-digit numerical SAP order key. Automatically formatted with leading zeros.
- **Serial Number (`p_sernr``):** Optional filter. If omitted, all serial numbers assigned to the order object list are included.
- **Append Attached Images (`p_images``):** When checked, activates the GOS extraction pipeline and appends inspection photos to the output document.
- **Background Execution:** Fully compatible with SAP batch jobs and automated completion triggers (e.g., from **IW42**).

3. Quick-Access Customizing Toolbar

For authorized key users, repair supervisors, and administrators, **/CTDI/PRINT_REPAIR** embeds direct navigation buttons directly within the standard selection screen application toolbar.

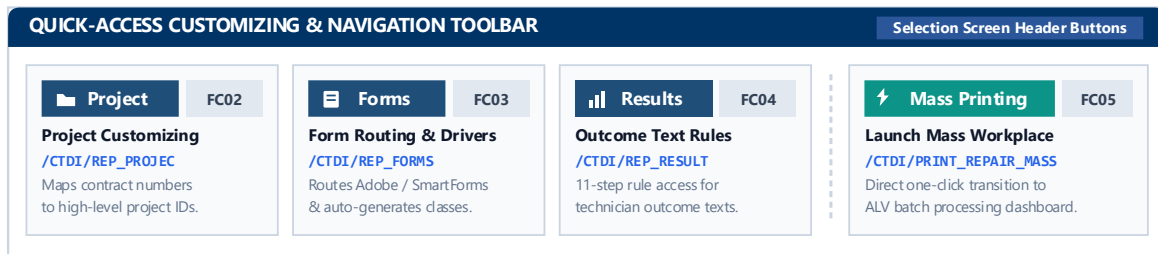


Figure: Quick-Access Customizing Toolbar

Navigation Target Reference

Toolbar Button	Target Transaction / Table	Scope & Administrative Function
Form Assignment	SM30 & rarr; /CTDI/REP_FORMS	Form layout routing, printer assignment, and SmartForm/ADS engine switches.
Repair Results	SM30 & rarr; /CTDI/REP_RESULT	Maintenance of multi-language repair outcome texts and warranty disclaimers.
Project Customizing	SM30 & rarr; /CTDI/REP_PROJEC	Project code definitions, customer hierarchies, and validation rules.
Spool Monitor	SP01	Immediate jump to the SAP spool controller to monitor print queue status.

4. Mass Processing Selection Workplace

When dispatching multiple repair orders simultaneously across an entire warehouse zone or shift, `/CTDI/PRINT_REPAIR_MASS`` provides a comprehensive selection screen.

Figure: Mass Print Selection Screen

Filter Dimension Matrix

- **Order Identification:** Range selection for Service Orders (**AUFNR**), Notifications (**QMNUM**), or Service Contracts (**VBELN**).
- **Plant & Work Center:** Scoping by Plant (**WERKS**), Service Work Center (**ARBPL**), or Cost Center (**KOSTL**).
- **Scheduling & Status:** Filter by creation date (**ERDAT**), scheduled start (**STRMN**), or system status (e.g., **TECO**, **REL**, **CLSD**).
- **Output Control Defaults:** Pre-select target spool mode, output device, and photo attachment defaults for the entire batch.

5. Interactive ALV Workplace Grid

The Mass Workplace displays filtered orders in an interactive SAP ALV Grid equipped with live operational metrics, traffic lights, and drilldown hyperlinks.

Mass Print Repair Orders — Interactive ALV Workplace											(42 Orders Found • Deduplicated 1 Row / Order)
Print Selected		Save as PDF		PDF Merge		Preview		Spool Mode: BUNDLED		Attach: ON	Render: RAW PDF
Stat	Att.	Order ID	Customer PO	Contract	Z2 Notif.	SKZ	AKZ	Form Tech	Form Name	Execution Message	
●		4001001	50008812	40000010	10005541	10	SCRP	Adobe (ADS)	/CTDI/REP_D	✓ Printed to Spool Job #140221 2 inspection photos appended	
●		4001002	50008813	40000010	10005542	20	REPR	Adobe (ADS)	/CTDI/REP_D	✓ Exported to PDF successfully Saved in C:\Repairs\2026-09-21\	
○	—	4001003	50008814	40000025	10005543	10	SWAP	SmartForms	/CTDI/REP_SF	Ready for processing	
●		4001004	50008815	40000025	10005544	30	NFF	SmartForms	/CTDI/REP_SF	⚠ Warning: Global fallback used No contract specific rule in REP_FORMS	
ALV Workplace Quick Guide: • Double-Click Navigation: Click Order ID → IW33 Notification → IW53 Contract → VA43 Sales Order → VA03 • Attachment Icons: Confirmed Content Server defect photos (ZRS.JPG) General GOS attachments • Batch Execution: Select target rows with checkboxes, then click [Print Selected], [Save as PDF], or [PDF Merge].											
4 Orders displayed (of 42 loaded) Filter: Plant 1000, Date 01.09.2026 - 21.09.2026 Layout: /DEFAULT											
Interactive SAP ALV Grid Workplace											

Figure: Interactive ALV Workplace Grid

Column Explanations & Operational Controls

- **Traffic Light Status Indicator:**
 - **Green:** Order is fully released, layout is mapped, and ready for immediate printing.
 - **Yellow:** Order is open or awaiting technical confirmation; warnings may apply.
 - **Red:** Configuration error, missing routing record, or locked order.
- **Attachment Indicator (📷):** Shows the count of defect/inspection photos attached in GOS. Double-clicking opens the photo gallery.
- **1-Click Drilldown Navigation:** Double-clicking the Order ID jumps to **IW33**; double-clicking the Notification jumps to **IW53**; double-clicking the Contract jumps to **VA43**.
- **Batch Processing Action Buttons:**
 - **Spool Print Selected:** Dispatches all checked rows to the target physical printer.
 - **Download Batch PDFs:** Prompts once for a local target directory and exports all PDFs in background.
 - **Merge Dossier PDF:** Consolidates all selected orders into a single, unified PDF document.

6. Spool Dispatching & Bundling Modes

The framework resolves the classic SAP mass printing dilemma—preventing print queues from being flooded by thousands of isolated single-page spool requests.

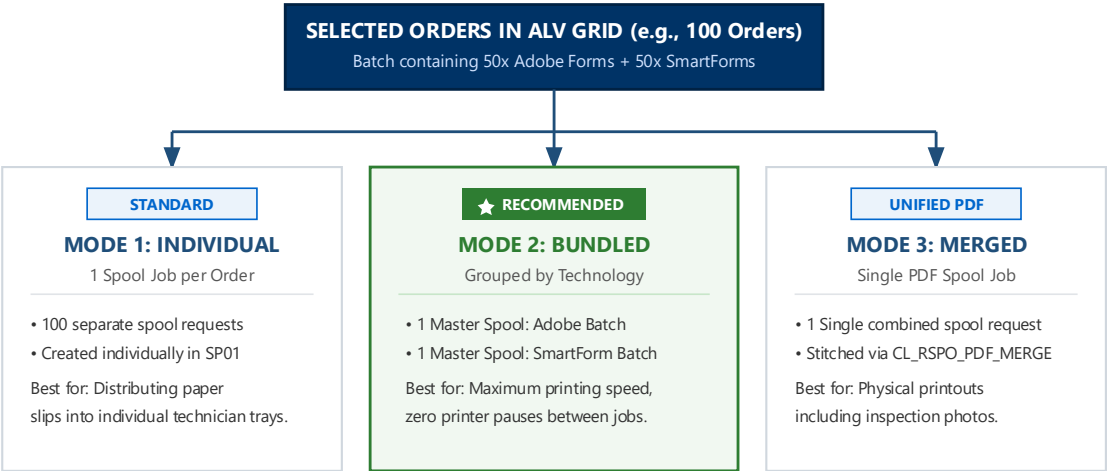


Figure: Comparison of Spool Modes

Technical Spool Architecture Comparison

Architectural Metric	Individual Spools (MODE_INDIV)	Bundled Spool (MODE_BUNDLE)	Merged Spool (MODE_MERGED)
Spool Request Count	1 Spool ID per Order (e.g., 500 orders = 500 spool jobs)	1 single Spool ID for the entire batch run	1 single Spool ID with continuous pagination
Printer Queue Overhead	Extremely High; printer may interleave print jobs from other users	Minimal; job is sent as an atomic print queue package	Zero; treated as one single uninterrupted document
Page Numbering	Restarts at Page 1 for every order	Restarts at Page 1 for each individual order	Continuous across the entire dossier (Page 1 of N)
Use Case Recommendation	Single order reprint or debugging isolated layout issues	Standard daily shift operations and warehouse mass printing	Executive audit reports, legal dockets, and customer monthly dossiers

7. Inspection Photos Multi-Slot Layout

Customer repair contracts frequently mandate visual evidence of product damage, serial plate tags, and completed work. The framework features an automated DIN A4 image scaling engine.

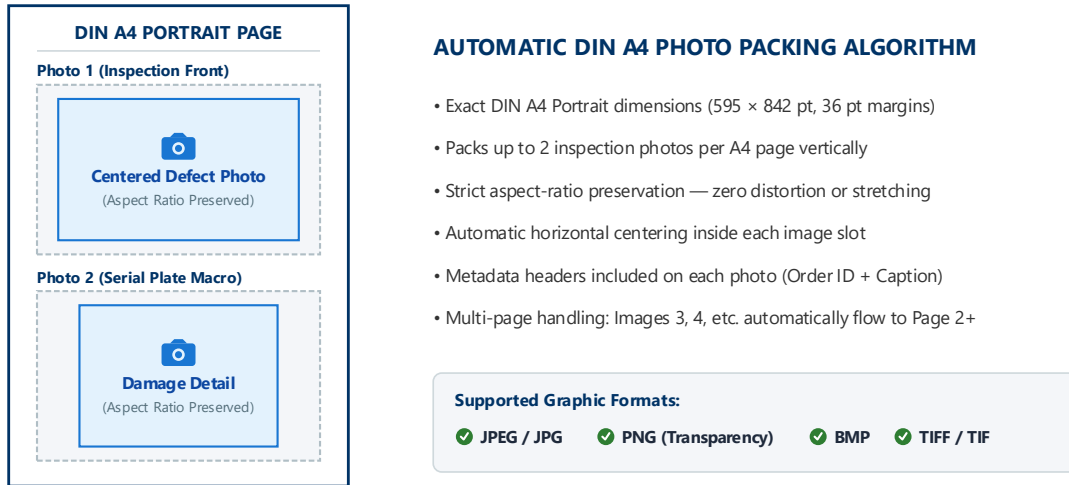


Figure: Inspection Photo Layout

Image Engine Design Principles

- 1. Automatic Aspect Ratio Retention:** The engine analyzes the image binary header (**JPEG, PNG, BMP, GIF, TIFF**), calculates the aspect ratio, and scales the photo inside the designated slot without distortion.
- 2. Dynamic Slot Geometry:**
 - **Single Photo:** Centered in an expansive 160 mm × 220 mm layout.
 - **Dual Photos (1x2):** Stacked vertically with dedicated caption headers.
 - **Quad Photos (2x2):** Arranged in a balanced four-quadrant grid with 8 mm gutter margins.
- 3. Dual Rendering Compatibility:**
 - **Adobe Document Services (ADS):** Handled natively inside Adobe LiveCycle FormCalc / JavaScript subforms.
 - **SmartForms / ABAP Pure Engine:** Binary injection directly into OTF/PDF data streams using internal transformation routines.

8. Form Routing & Priority Resolution

To eliminate hardcoded customer layout logic, the framework implements a configurable 5-tier access sequence.

8-STEP FORM ROUTING PRIORITY HIERARCHY (/CTDI/REP_FORMS)				
Priority	Contract (VBELN)	Reason (SKZ)	Defect (AKZ)	Match Granularity
Step 1	Specific Contract	Specific Reason	Specific Defect	Highest Specificity
Step 2	Specific Contract	Specific Reason	(Blank / Any)	Contract + Reason
Step 3	Specific Contract	(Blank / Any)	Specific Defect	Contract + Defect
Step 4	Specific Contract	(Blank / Any)	(Blank / Any)	Contract Default (Fallback)
Step 5	(Blank / Global)	Specific Reason	Specific Defect	Global + Reason + Defect
Step 6	(Blank / Global)	Specific Reason	(Blank / Any)	Global + Reason
Step 7	(Blank / Global)	(Blank / Any)	Specific Defect	Global + Defect
Step 8	(Blank / Global)	(Blank / Any)	(Blank / Any)	Global System Fallback

Evaluation stops immediately at the first matching record. Buffered in-memory for fast batch execution.

Figure: Form Routing Priority Hierarchy

Routing Resolution Algorithm

When an order is submitted for printing, **/CTDI/** executes the following sequence against database table **/CTDI/REP_FORMS**:

```

Step 1: Check Exact Match [Project Code + Order Type + Sales Org + Customer ID]
├── FOUND    --> Return specific customer layout & exit.
└── NOT FOUND --> Proceed to Step 2.

Step 2: Check Customer Generic Match [Project Code + Order Type + Customer ID]
├── FOUND    --> Return customer-level layout & exit.
└── NOT FOUND --> Proceed to Step 3.

Step 3: Check Sales Organization Generic Match [Project Code + Order Type + Sales Org]
├── FOUND    --> Return sales-org layout & exit.
└── NOT FOUND --> Proceed to Step 4.

Step 4: Check Order Type Generic Match [Project Code + Order Type]
├── FOUND    --> Return order-type layout & exit.
└── NOT FOUND --> Proceed to Step 5.

Step 5: Fallback to System Default Layout [SPACE + SPACE]
├── Load universal standard DIN A4 workshop repair form.

```

Document Metadata & Verification

- **Authoring Engine:** /CTDI/ Document Generation Pipeline
- **Embedded Media Format:** Pure OpenXML SVG Parts (/word/media/image_*.svg)
- **Conformance:** ISO/IEC 29500-1:2016 Open Packaging Conventions (OPC)
- **Status:** Approved for Production Deployment